

Apply to be a Science Corps Fellow - Entry #3402

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Gender
Female
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British
When did you or do you expect to complete your PhD?
I expect to complete my PhD in the summer of 2026, likely around August. I am therefore interested in a fellowship beginning in November 2026.
Please tell us briefly about your university education (institution, city, country, year)
I earned my MChem in Chemistry (1st Class) from St Hugh's College (2018–2022), University of Oxford, United Kingdom. I completed my master's project in the group of Professor Andrew Goodwin FRS, with the thesis title, "Cooperative Jahn–Teller order in Prussian blue analogues". I began my DPhil (PhD) in Inorganic Chemistry in 2022 at Hertford College, University of Oxford, under the supervision of Professor Andrew Goodwin FRS (University of Oxford) and Professor David Keen (ISIS Neutron and Muon Source, UK). I am aiming to complete my DPhil studies around August 2026. During my DPhil, I have been using both experimental and computational techniques to understand local structure problems in solid-state materials, predominantly in Prussian blue analogues, which have emerged as promising battery materials. I am interested in disordered materials, particularly where disorder is not random but correlated, and how the interplay of different distortions can drive useful material properties.
Why would you like to be a Science Corps Fellow?
I am passionate about science. Scientific research can have immeasurable impacts on many lives, solving complex global issues using our understanding of the world. A good science education is therefore critical, as it fosters logical and critical thinking, evidence-based decision making, as well as curiosity and creativity. Science education should be open to all. I am therefore passionate about open, clear, and engaging science communication and using the power of education to build a more equal and inclusive world. I hope that a fellowship with Science Corps would

help me to contribute towards these aims.

I am curious about the sheer diversity and range of materials that surround us every day. Through understanding the nature of materials, the ability to exploit their properties to make a positive impact on a wide range of challenges faced by society, including renewable and sustainable energy, is what draws me to materials chemistry. From a more selfish curiosity, I am fascinated by the concept of correlated disorder and the interplay of different distortions and how they can be exploited to improve material function. I believe this love comes from my desire to find links, whether it be in numerical data or visual patterns, something a strong science education has allowed me to develop. Doing science research satisfies my desire to understand the world around me, whilst seeing the real-world applications of what I learn. I hope that my enthusiasm for science might encourage others to explore their own interests in science. I believe that mentors have the power to create positive learning experiences that can spark a curiosity and love for science that can transform lives. I would relish the opportunity to develop a programme with Science Corps that teaches scientific content in an engaging manner, whilst highlighting links to real-world applications. I believe that there is no one way to learn and that students should have the opportunity to explore different learning media and find what works for them.

In an interdisciplinary field such as materials chemistry, it is hugely beneficial for people with different expertise to look at the same problem from different perspectives. I strongly believe that a culture of accessibility, creativity, and collegiately leads to good science. I hope that gaining international experience through Science Corps, sharing our different approaches to teaching, learning, research, and life, across both cultural and academic contexts, will help broaden my perspective, both professionally and personally.

A strong science foundation is extremely important in life, not just for solving global issues, but also for ensuring students are equipped with the vital skills, knowledge, and creativity needed to build a better future. I therefore wish to improve my skills as a science educator with Science Corps, whilst giving back and hopefully convincing students that there is a place for everyone in science.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

From academic opportunities, such as teaching chemistry, to extra-curricular activities, such as captaining the Oxford University Women's Cricket Club, I have been able to work on skills, such as communication and leadership, that would be important for a Science Corps Fellow.

Tutoring undergraduate inorganic chemistry at St Catherine's College, Oxford, and supervising master's students within my research group, has been incredibly rewarding. Whilst teaching can be challenging at times, requiring patience and persistence, being able to guide students to think independently and ask insightful questions, has shown me how science can empower people with logic, thoughtfulness, and respect for other people's opinions. I prepare teaching materials, give small-group tutorials, create tests, and mark work, but a large part of my role is as a mentor, to share my enthusiasm and love of science whilst encouraging others in their journey. Self-doubt is common in science and having a mentor can be crucial to building the confidence someone needs to see that there is a place for them in science. To give fair undergraduate chemistry admission interviews, I completed safeguarding and contextual data training, including unconscious bias awareness. I hope that what I have learnt from my experiences of teaching so far would give me a good foundation for a successful Science Corps Fellowship.

In terms of both learning and teaching science, I believe in the power of a holistic approach, making use of different media to suit different learning styles. Solid-state chemists use a combination of maths and different visual representations to understand structure and symmetry. In my research, I use a range of complementary methods, both computational and experimental, to build a more complete picture of the systems I am studying. In our lab, we also make use of 3D printing to understand perhaps counterintuitive mechanical properties such as negative thermal expansion. I believe that if science educators can make science more engaging through using different learning media, we will be more likely to make science fun and accessible and encourage more students to pursue a career in science. I hope that my experience of using different learning media will allow me to help students engage with science.

I am also passionate about science communication, which is key to making scientific knowledge accessible to all. I enjoy the challenge of creating visually engaging prize-winning posters, translating complex information into easier-to-understand content. I am grateful for the opportunities I have had to engage in science discussions with people from different scientific and cultural backgrounds at international conferences. Our research group goes on a yearly retreat abroad, which I always find to be my most productive week of the year. I enjoy travelling, also solo, and the enrichment and new perspective it brings.

Where do you imagine yourself professionally in the future? Check all that apply.

Scientist in academia

In science education

In science outreach

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

In the near future, I would like to do a postdoctoral research position in materials chemistry, learning new and complementary skills whilst leveraging my current skill set to investigate and solve problems in either the same or a related area. It would be valuable to see how others use similar techniques to solve problems in different areas, or equally, to see how the same problem can be approached from different angles. I hope that doing a Science Corps Fellowship soon after completing my PhD would give me the time and space to reflect on what I enjoyed most during my PhD studies, to help guide me towards finding my niche. I would also like to build my confidence, independence, and grow my network. Teaching is also something I very much enjoy, so I would also love to carry on teaching, whether it be helping guide research students at the start of their journey or teaching classes. Equally, I still have a thirst for knowledge and a desire to keep learning.

In the long-term future, I might consider starting a research group of my own. I find a uniquely appealing aspect of academia to be the chance to make positive contributions to society through science advancement, whilst also having the opportunity to continually explore and learn new things, whether it be through knowledge, technical or interpersonal skills. I would want my group to have a supportive and relaxed environment, with an emphasis on diverse thinking where everyone and anyone regardless of background feels welcomed and valued. I am passionate about science education, which is also at the heart of academia, so I would enjoy the opportunity to mentor students, teach, lecture, and give back to the community.

I care deeply about science communication and its ability to improve scientific literacy, build trust, and encourage participation in science. A venture I would like to explore would be building a website to boost science engagement, adding a 'human' element that makes science seem more relatable to non-experts. I would include a blog, touching on subjects such as the life of a scientist in academia with contributions from colleagues, with the aim of providing honest perspectives to dispel myths about scientists. I hope that sharing our unique journeys would help young people see a path for themselves in science. I would like to create engaging online resources using different media, such as animations, videos, podcasts, interactive tools, and articles, to create content that appeals to a wide range of audiences; I believe that a Science Corps Fellowship would give me a good foundation in this regard. I want to share the beauty of science and the science behind beauty. I have been very lucky to share in interesting discussions with visual artists during my PhD studies, from exploring topics of how to present ideas visually to aid understanding, to exploring (dis)order and patterns in art. I very much hope that collaboration will be at the heart of all my future work.

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